```
FUNCTION convertInfix(expression, mode):

    IF mode == "PREFIX":
        expression ← REVERSE(expression)
        SWAP all '(' with ')' and ')' with '('

    Stack ← EMPTY
    output ← ""

    FOR each token IN expression:

        IF token is OPERAND:
            output ← output + token

        ELSE IF token == '(':
            PUSH token TO Stack

        ELSE IF token == ')':
            WHILE Stack.TOP != '(':
                output ← output + POP(Stack)
            POP(Stack)

        ELSE:
            WHILE Stack NOT EMPTY AND PRECEDENCE(Stack.TOP) >= PRECEDENCE(token):
                output ← output + POP(Stack)
```

```
        PUSH token TO Stack

    END FOR

    WHILE Stack NOT EMPTY:
        output ← output + POP(Stack)

    IF mode == "PREFIX":
        RETURN REVERSE(output)

    RETURN output

END FUNCTION



FUNCTION PRECEDENCE(op):
    IF op == '^'       RETURN 3
    IF op IN {*, /}     RETURN 2
    IF op IN {+, -}     RETURN 1
    RETURN 0
END FUNCTION
```